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P. A. M. Dirac - Session II

May 6, 1963

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Location: Cambridge, England

Transcript version date: December 18, 2024

DOI: <https://doi.org/10.1063/nbla.hvrs.gjyc>

Abstract:

Part of the Archives for the History of Quantum Physics oral history collection, which includes tapes and transcripts of oral history interviews conducted with circa 100 atomic and quantum physicists. Subjects discuss their family backgrounds, how they became interested in physics, their educations, people who influenced them, their careers including social influences on the conditions of research, and the state of atomic, nuclear, and quantum physics during the period in which they worked. Discussions of scientific matters relate to work that was done between approximately 1900 and 1930, with an emphasis on the discovery and interpretations of quantum mechanics in the 1920s. Also prominently mentioned are: Niels Henrik David Bohr, Max Born, Boyland, Louis de Broglie, Johannes Martinus Burgers, Paul Ehrenfest, Ralph Fowler, Peter Fraser, Werner Heisenberg, Ernst Pascual Jordan, Cornelius Lanczos, Edward Arthur Milne, Wolfgang Pauli, David Robertson, Ernest Rutherford, Erwin Schrödinger, John Joseph Thomson, Hermann Weyl; University of Cambridge, Delta Squared V Club, Kapitsa Club, Kbenhavns~Universitet, Merchant Venturer's School in Bristol, University of Bristol, and Universität Göttingen.

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Kuhn:

I'd really like to get a clearer idea of what you may actually have covered In school science and mathematics, then later at Birmingham, and at Cambridge.

Dirac:

In the first place it was Bristol, not Birmingham. I went to a very good science school in Bristol, a Merchant Venture's School. They just concentrated on science and on modern languages. They didn't teach Latin or any kind of classics at all. Just a little bit of history and geography. I didn't like the arts side of it. I feel I was very lucky to get to this good school. The school shared the same building with the technical college. It's a school in the daytime and a technical college in the evening so all the laboratories of the technical college were available to the school. We had very good laboratory facilities. Then another thing which helped me was that as soon as I went to this secondary school the war started. That meant that the older boys were taken away, called up for war work, the higher classes were empty and they pushed on the younger boys as quickly as they could. So I got pushed on ahead and learned science earlier than I would otherwise have done.

Kuhn:

Was it quite unusual to have no classical languages in a school of this sort?

Dirac:

Well, most secondary schools would teach Latin. This one didn't.

Kuhn:

Did. that mean that it was designed to prepare people particularly for technical schools?

Dirac:

Well, it was meant to give a good scientific education. I don't know of any school now which concentrates so much on science as that one did.

Kuhn:

But would it mean, for example, that students who had graduated from there generally could not go to Oxford or Cambridge without Latin in that day?

Dirac:

They would have to take Latin as a separate subject; not everyone had to go into it.

Kuhn:

What sort of schools would most of them have gone to?

Dirac:

. . . They would go to Bristol University or other modern universities.

Kuhn:

Would. they be particularly likely to go into the sciences or engineering?

Dirac:

They would from that school.

Kuhn:

Can you tell me more about what subjects you would actually have prepared there in the sciences?

Dirac:

No biological sciences. That hadn't come into the ordinary school work then, but (we had) physics, chemistry, mathematics; mathematics was divided into algebra, geometry, trigonometry. Of course I was soon working independently in mathematics — independently of the class. I was just given books to read which I worked through by myself.

Kuhn:

Do you remember what books?

Dirac:

Well, there was Edwards Calculus, differential calculus and Integral calculus. I think it was Hall and (Knight's) Geometry, I'm not sure. I don't remember very clearly what the books were. I know we did not stick to Euclid, We had more modern methods of geometry than Euclid. There were mainly the science courses and courses in English, French and German. We had three hours a week in each of these subjects.

Kuhn:

How much would have been in the physics course?

Dirac:

There may have been three hours a week of lecturing and one afternoon of practical work.

Kuhn:

Just for the one year?

Dirac:

I think all the time. Chemistry also I started quite early because of being pushed on on account of the war. I remember my chemistry master, Mr. Boyland, believed in teaching chemistry in the modern way and he introduced atoms and chemical equations very early, in the course. I think in almost my first lecture in chemistry he introduced me to these things. We learned about atomic weights and we did not learn about equivalent weights. I think even now in school in chemistry they spend a lot of time on equivalent weights and only go on to atomic weights later on. I know my daughter was learning quite a lot about equivalent weights.

Kuhn:

Well, I think I had them both but I think I had atomic weights before I had equivalent weights.

Dirac:

Yes. I hardly ever heard of equivalent weights. I hardly knew what they meant. Well, do you have any other questions?

Kuhn:

What sort of subjects in physics would you actually have gotten - through using how much of the accompanying mathematics either by yourself or in the course?

Dirac:

It was only in mathematics that I worked by myself, in the other subjects I worked in the

class. I don't remember. I suppose they were just the usual things. Heat, and light and sound. It was spread over four years, you see. I had four years in that school.

Kuhn:

Where you worked by yourself in mathematics in that school, did you then come back and talk to the teacher about this or did you simply tell him that you had read the books?

Dirac:

I expect I talked to him a bit. I think I was mostly on my own. I was just allowed to read what interested me. I took the same exams as the rest of the class. They were no problem to me.

Kuhn:

Were you reading at this time by yourself in other subjects? I don't necessarily mean academic subjects. Did you read novels? Or poetry?

Dirac:

I read some novels. In fact, one was supposed to do it in connection with the English lessons. I was always a slow reader and I didn't like poetry. In fact, I never understood good poetry.

Kuhn:

Well, this was not something you went on with, particularly then on your own as you did with the math.

Dirac:

No, no. My whole interest was on the scientific side. I was really very ignorant of matters outside my schoolwork.

Kuhn:

I think you were also quite interested at times in economics.

Dirac:

Not at that stage.

Kuhn:

When did that begin?

Dirac:

I was never very much interested in economics. Who said that I was?

Kuhn:

I'm not sure. You gave an economic example in your Nobel address, for example. Someone spoke also of an occasion when you had talked on, I think, an economic, perhaps a social, problem when you might at least equally well have talked about matters more strictly scientific.

Dirac:

It must have been much later because... Certainly not in that period at all. I took no interest in political things either.

Kuhn:

Now let me ask you about subjects and courses at Bristol.

Dirac:

In the school or university?

Kuhn:

Now I mean at the university, assuming that we have done this now for the pre-university.

Dirac:

Well, I was in electrical engineering at Bristol and that was in the same building again as this school and the technical college. They were all together. The faculty in engineering was quite small then. They were all housed in the same building so that when I went to the university I continued to go to the same building. To some extent I had the same teachers. I specialized in electrical engineering and I was mainly influenced by the professor of electrical engineering there called David Robertson. He was extremely methodical. He taught both theoretical and practical work. I remember how methodical he was with the practical work, impressing upon people the need to arrange things so as to avoid the possibility of accidents. He was paralyzed from polio and he rode around

everywhere. He was paralyzed in his legs and he went around everywhere in a wheelchair. So he had to arrange his own life methodically.

Kuhn:

This made no great problems for him in teaching the practical work?

Dirac:

Well, he could go around in his wheelchair and see what everyone was doing and explain things to them.

Kuhn:

How theoretical was the teaching for electrical engineering of electromagnetic theory or how much of electromagnetic theory as such did you get in that curriculum?

Dirac:

Well, engineers just concentrate on the equations that give the right answers. They don't want to know the reasons for those equations. I learned the equations and how to use them — all about inductance, capacitance and so on, how to work out electrical circuits and the rules for winding electrical motors and dynamos. I found it was quite interesting mathematics in some of these rules. I am grateful to David Robertson for explaining these things in a way which did show the mathematical beauty of some of the things which one had to calculate.

Kuhn:

Did one go as deep as Maxwell's equation?

Dirac:

We didn't do electrical waves at all. Wireless didn't exist as a regular thing.

Kuhn:

So this was really for power engineering.

Dirac:

Yes, yes.

Kuhn:

Did you learn to do what I once learned to call Heaviside calculus?

Dirac:

I did a bit of that with solving differential equations, linear differential equations with constant coefficients. I was just taught it as a rule that worked. I couldn't explain very well why they did work, but still they did work and gave the answer that you wanted, and that was enough to satisfy engineers.

Kuhn:

Were you impressed with that?

Dirac:

There is some sort of magic about it, yes. It was strange how you could get the answers out.

Kuhn:

I had just a taste of it and didn't really get back to it until I began to learn a little bit about operator techniques and delta functions. You had these in the reverse order and I wondered whether they had any kinship.

Dirac:

Well, I was just first taught the way to — the rules which one had to use to get the right result.

Kuhn:

I've got no notion how specialized or how quickly specialized the electrical engineering curriculum would have been. Did you do simply electricity or did you...

Dirac:

Well, there was some amount of general engineering, for instance, testing materials, pulling things until they break and calculating the breaking stress, and things like that. We had quite a good deal about calculating stresses in structures.

Kuhn:

Would you have had a general dynamics course along with this?

Dirac:

Yes, yes. One had to know about the dynamics. There was quite a good deal of general engineering in the work.

Kuhn:

Was the approach to the scientific problems generally rather different from what one would have had if one had been in physics or was this quite a lot the same?

Dirac:

I think it was different in the sense that they just concentrated on learning equations that give the right answer. Then you could proceed to work out the answer with slide rules.

Kuhn:

Were you at all impelled yourself at that point to push back to a more basic source for these formulas?

Dirac:

I don't think I felt impelled to it. I think at that time I wasn't really doing much inquiring; it was just absorbing the knowledge which was given to me. I think it was! probably that sort of training that first gave me the idea of a delta function because when you think of loads in engineering structures, sometimes you have a distributed load and sometimes you have a concentrated load at the point. Well, it is essentially the same whether you have a concentrated load or a distributed one but you use somewhat different equations in the two cases. Essentially it's only to unify these two things which sort of led to the delta function.

Kuhn:

Do you think you did try to unify them at that time?

Dirac:

Not at that time. Oh, no. Not at that time at all. But I just felt that they were essentially the same even though the treatment was different. There was a lot to be done just calculating stresses and things like that. Maybe the stress on a whirling shaft, and I found that some of the mathematics was quite pretty even though the whole work ± 3

approximate. I think I owe a lot to my engineering training because it did teach me to tolerate approximations. Previously to that I thought any kind of an approximation was really intolerable. One should just concentrate on exact equations all the time. Then I got the idea that in the actual world all our equations are only approximate. We must just tend to greater and greater accuracy. In spite of the equation's being approximate they can be beautiful.

Kuhn:

When you say approximate here, I think people often use the same term in two rather different ways. This may mean that they give approximate results in the sense that engineering equations do. It may mean that the whole physical theory with which the equations are involved is approximate in the sense that it's an approach to what's really there.

Dirac:

Well, it's just neglecting a lot of factors. I meant it in that sense. The actual situation is far too complicated, you have to neglect a lot of factors.

Kuhn:

Would you say that one gets the Bohr atom from quantum mechanics or that one gets to quantum mechanics from the Bohr atom by adding simply a factor that one has been ignoring previously, which would give a sense of approximation?

Dirac:

You mean by quantum mechanics the Schrödinger equation.

Kuhn:

The Schrödinger equation or the general -. I mean would that also be an example of an approximation in the sense that you have in mind?

Dirac:

Yes, it would be. Yes.

Kuhn:

Because clearly the logical relation of the earlier and later theory there is more complex; that is, it would be difficult to isolate a single — or group of extra things, that sort of

added on

Dirac:

I think it's very likely that all our equations are only approximate. Our present quantum theory is probably only an approximation to the improvement of the future. I feel that everything might be an approximation and this comes very largely from the engineering training.

Kuhn:

I think this question may arise again when we come back to some of the problems presented by quantum mechanics. You may be willing to elaborate it because I think most engineers' idea of what is involved in the approximation is that there may be a complete theory, an exact theory, which you must know in order to know what you're neglecting.

Dirac:

I don't think that's the engineering approach. I think they have a general feeling about what is important and what can be neglected. That would make a good engineer. Probably would make a good physicist also.

Kuhn:

But this is a sense of an approximation in which one doesn't ask except perhaps intuitively to know what the full series would be in order to justify dropping the term.

Dirac:

Yes, yes. You just rely on your general intuition. That's certainly so for engineers.

Kuhn:

Very good. In addition, were you doing anything over and above your engineering studies at this point as you had gone ahead by yourself with math at school?

Dirac:

I continued to go ahead with math and I was interested in relativity. You see, just at that time relativity sort of burst on the world. That was 1918 when the war ended and relativity had a tremendous effect on the general public. Papers were full of articles on relativity and magazines also. Every one was talking about relativity. It was a

new philosophy and it produced more excitement I think than science has ever produced before or since. Of course, engineering students were also interested in it and talked about it. And during in my later period I worked on it. I told you that I went to Professor Broad's lectures. remember I was also interested in other things, philosophy and logic. I got a copy of Mill's from the library and read that all through. I found it pretty dull, but still I stuck to it, and tried to understand these things. I thought at that time that maybe philosophy was important. Since then the field of philosophy has terribly declined. I feel that philosophy will never lead to important discoveries. It's just a way of talking about discoveries which have already been made.

Kuhn:

What else did you read in philosophy?

Dirac:

I can't remember reading any other books.

Kuhn:

Were there philosophical problems that you talked to people about or worried about yourself? Bohr, for instance, talked a good deal about how concerned he had been at one time with the problem of the freedom of the will.

Dirac:

I never talked to people about it. I thought about it, but found that sort of problem insoluble. I did think a lot about these things but I never came to any definite conclusions.

Kuhn:

Do you remember what the particular problems were?

Dirac:

I think I told you that one time I thought that space and time might be connected and that you might have to rotate space and time axes together. But I only knew about Euclidean space then and, of course it won't work with Euclidean space. So I had to give up with that.

Kuhn:

Were some of the more usual philosophical problems like the freedom of the will, the reality of other minds?

Dirac:

I thought about those things, yes. Can't remember anything definite that this thinking led to.

Kuhn:

You spoke, when you talked before about relativity, about having gotten hold of Eddington's book. Were there other books at that point that you were particularly working in yourself?

Dirac:

I don't think so, no. Eddington's book was the first mathematical book on relativity.

Kuhn:

Well, I've noticed when you get up to Cambridge you start publishing quite soon. Your second semester. Now clearly you've got a very good command of statistical mechanics. Did that all happen while you were there?

Dirac:

No. I might say that there was a good deal of controversy at that time between Eddington and Jeans and some of the students were very interested in that. One of the students that I was especially friendly with called Wilshire used to be very keen to read letters in Nature which Eddington and Jeans had written.

Kuhn:

This was a controversy over relativity?

Dirac:

I've forgotten what it was about. No, I think it was some astronomical thing. One can look it up. Wilshire was also interested in relativity and we had talks about that together.

Kuhn:

You said that you thought your choice of engineering was probably largely determined

by the fact that your brother had done this.

Dirac:

Yes, and also it was in the same building and a natural thing for students to go on to. I suppose I didn't know then that one could get one's living in pure science.

Kuhn:

Had you been tempted to do mathematics, for example? It was the thing you had been reading by yourself.

Dirac:

It was the thing which interested me most but I did not know that one could get a living by it. No one ever spoke to me about it. Engineering was something which one could earn one's living from 'and I knew that.

Kuhn:

Well, teaching mathematics, would you have thought of that as a way of making one's living?

Dirac:

I didn't like teaching very much. Of course, I thought I could have been a school teacher but I didn't like it.

Kuhn:

So you think probably that you considered mathematics as a possibility and rejected it because it meant school-teaching which you didn't want to do.

Dirac:

Yes.

Kuhn:

How did you feel about the practical work in engineering? You've spoken of the appeal of engineering approximations, but what about the manipulative, mechanical "soldering-side" of electrical engineering?

Dirac:

I don't suppose I was very good at it. On one long occasion I went to a factory to get some practical training in engineering work. They sent an unfavorable report on me to my professor, so I presume I didn't do very well.

Kuhn:

Did you enjoy the more manipulative part —.

Dirac:

I enjoyed it, yes, but I was extremely ignorant of anything which I hadn't learned in class. I was just lacking in., common sense, I suppose.

Kuhn:

Did you ever do experimental work after you made the transition to physics?

Dirac:

I did a very little at one time, yes.

Kuhn:

I ran onto a letter, which I think was printed in the Fermi volume, from Rutherford to Fermi in response to some news about the early neutron work. Rutherford congratulates Fermi on turning to experimental work and then says that you're also. doing some experimental work so there's hope for theoretical physics.

Dirac:

Yes. I did a little work on trying to separate isotopes by centrifugal methods.

Kuhn:

How did you core out with that? I know we're out of chronological order, but we're not likely to come back to this one by the logic of your paper.

Dirac:

Well, I was a close friend of Kapitza at that time and I think he interested me in it and persuaded me to take up some experimental work; I had the idea that maybe just the rotation of gases at high speed without any mechanical moving parts could be used for

separation of isotopes. I did some experiments on it and got a negligible amount of separation, but I got another effect which I wasn't expecting which was interesting, namely a thermal effect. I was able to produce something like a conjuring trick. I just showed three pipes, pumped in some air into one pipe; have the one pipe here branching out into two pipes and the air would come out of these two pipes at different temperatures, widely different temperatures.

Kuhn:

They were coming off symmetrically?

Dirac:

Well, I mean there were some asymmetry in the apparatus but the apparatus was so small that it was concealed- . I believe we had differences of about maybe 1000 centigrade.

Kuhn:

At what temperature?

Dirac:

The pressure was about six atmospheres. Maybe not quite that much; but below freezing point on one side, and the other side quite high. Comes entirely from the viscosity effects. The inner layers have a higher angular velocity than the outer ones and the viscosity transfers energy from the inner layers to the outer ones. The outer ones go out one side and the inner ones come out the other side and there's quite a big temperature difference.

Kuhn:

Did anything further come of that?

Dirac:

I tried to see whether there was any separation when one used a mixture of gases and it seemed to me the simplest way to try that would be to put some scent in it, the scent being some heavy gas, and see whether more scent comes out one side than the other but it turned out later that it was just due to the fact that one's sense of smell is very much keener at low temperatures than at high temperatures. The temperature effect quite dominated any other effect.

Kuhn:

When you say you had thought of mathematics and felt that there was not a living to be made there, was physics a career one would even think of at this time?

Dirac:

I never thought of it. No one ever spoke to me of it as a career. I did not come into contact with the physics and mathematics people at the university because they were in quite a different building a half a mile away. I never went to that building. I was entirely at this Merchant Venture's College.

Kuhn:

Do you mean to be taken literally when you suggest that its being in the same building was perhaps a major factor in the choice?

Dirac:

I think it was quite an important factor, yes. And the same people, same staff, to some extent. This David Robertson, for instance, who was my physics teacher in the school, I also —.

Kuhn:

So it was really because these were the subjects you enjoyed and you went right on with them.

Dirac:

Yes. With the same professor.

Kuhn:

I see that more clearly now. I'd taken it initially to mean, 'Why go a half mile away to another building?' But I see much more now the sense of continuity in a subject which you liked and were doing well with.

Dirac:

Yes. The people there rather took it for granted that I would continue in that way.

Kuhn:

Well, the last time you talked, you said that you had heard later that the mathematicians

were quite disappointed that you decided to do engineering,

Dirac:

Yes, yes.

Kuhn:

This sounds as though the mathematics was once a more live option, as though they might have either persuaded you to do it or you had talked about doing it or something.

Dirac:

They had heard about my good examination results and they hoped that I would specialize in mathematics.

Kuhn:

But they didn't try to persuade you?

Dirac:

I never heard anything of it. Perhaps it's going too far to say they never did anything, but if they did, it's skipped my mind. Well, going back to this centrifugal force, there's one further thing to add. When Kapitza was detained in Russia, I rather stopped. I didn't have enough enthusiasm to carry on without him and so nothing further was done on it until the war came and people wanted to separate isotopes of uranium and the subject was taken up again some modified apparatus was experimented with in Oxford. I went to Oxford from time to time to talk with the people there who were doing experiments. There is a possibility of getting small effects from the separation of gases by rotations which are produced without any mechanical moving parts.

Kuhn:

It never, I take it, was done on a major scale?

Dirac:

No, I don't think it was good enough to compete with the diffusion method. That was all the experimental work that I did as a physicist. Of course, I had to do a lot during my training in school and in the university.

Kuhn:

You said before that when you first went into the school of mathematics at Bristol, you really started first by going back to engineering school because you had not been able to get an engineering job in the depression.

Dirac:

That is so, yes. This professor, David Robertson, told me I shouldn't waste my time hanging around doing nothing and that I should go back and do some research. —He started me off on a problem with Stroboscopes, and after a few weeks there the mathematics people asked me to go and study mathematics with free tuition.

Kuhn:

Well, this is clearly the beginning of your way back or your way to physics. But I take it earlier if you'd suddenly heard of an engineering job, you would have taken it.

Dirac:

Yes.

Kuhn:

Is there anything identifiable as a point of decision?

Dirac:

Well, I really went over to mathematics in that period, not physics.

Kuhn:

Yes. But is there a point at which it suddenly becomes clear to you that you're not going to go to an engineering job if it does become available?

Dirac:

No, I don't think there's any definite occasion like that. I stopped applying for jobs when I went to study the mathematics at Bristol.

Kuhn:

And so far as you remember you never did open that up again.

Dirac:

No, I never applied again. Well, having started on a course, I wanted to finish it.

Kuhn:

That was a two-year course from the beginning, was it?

Dirac:

Yes. It would normally be a three-year course but with my previous training they let me off a year. There was another choice that had to be made, because in the first year of that two-year course one did pure and applied mathematics together, half and half. The second year one had to specialize in either pure or applied. So I didn't really know which to choose. I didn't mind very much between then. There were just two students doing the honors course then, a girl, called Miss Dent, and me. The professors, of course, wanted both of us to choose the sane option because it would mean only half as much teaching for them to do. But Miss Dent was very definite that she wanted to do applied and that was why I did applied mathematics rather than pure. Again, it's rather fortuitous that I got onto the path of applied mathematics.

Kuhn:

What subjects did you do — this would be a course that met every day?

Dirac:

Oh, yes. There were lectures every day, I think. Well, when I was doing both pure and applied, I was very much influenced by Peter Fraser whom I told you about. I learned projective geometry from him and I found it very interesting. It was really a fascinating subject. I found an article about Fraser (looks for article). You might like to look at it. You can take it along and read it.

Kuhn:

Thank you. Do you remember where it's from?

Dirac:

Hodge wrote it, and you could ask him. Professor Hodge is in Cambridge.

Kuhn:

Well, I shall take that along and have a look at it and bring it back. I'm very glad to see it.

Dirac:

I'd say he was the best teacher I ever had.

Kuhn:

Did you have him just in this one course?

Dirac:

No, he taught other things also, rigorous foundations in mathematics, how to differentiate and integrate with rigor. I previously had just learned the rules well enough for engineering and I found it rather hard to appreciate that rigor was needed. It seemed to me that when you were confident that a certain method gave the right answer, you didn't have to bother about rigor. In fact, I still feel rather that way. But Fraser did a very good job of persuading one of the need of rigor.

Kuhn:

Did any of your math get you more toward algebra?

Dirac:

By algebra do you mean sort of noncommutative stuff?

Kuhn:

Well, at least something more - just a class of variables that don't necessarily have to be numbers.

Dirac:

I remember reading about quaternion's by myself. I never had it in class but I did get hold of an old book on quaternion's.

Kuhn:

Do you have any notion what that would have been

Dirac:

Well, there weren't very many books on quaternion's. Who was it, Thomson, Tait?

Kuhn:

There are some quaternion's in Thomson and Tait.

Dirac:

This is a book entirely on quaternion's. A big thick book,

Kuhn:

That thick book on quaternion's would not be Thomson and Tait.

Dirac:

There weren't very many books on quaternion's in those days. They talked about tensors and versors. It was an old-fashioned book.

Kuhn:

Good. I'm much interested. You've mentioned Thomson and Tait and knowing something about quaternion's from that last time we talked —.

Dirac:

Maybe Thomson and Tait is the wrong book and I read something else.

Kuhn:

But you do remember there being a book on quaternion's by themselves. How did you like that?

Dirac:

I liked it in some ways, but I didn't appreciate it to the full extent because the authors did talk about tensors and versors for the scalar and vector parts and they rather separated the scalar and vector parts too much, instead of thinking of the two together as one entity. They put the wrong emphasis on it. So I didn't appreciate it as much as I ought to have done if I'd had a better book.

Kuhn:

This connects with something you said last time that puzzled me a good deal, because I

think in talking about the projective geometry with Fraser, you said that ever since then your own approach has been largely geometrical.

Dirac:

Yes.

Kuhn:

I was somewhat puzzled about this because I would myself have thought of your approach as being very often algebraic, but I probably wouldn't have even raised this except that as you may know Oppenheimer arrived in Copenhagen just before I left. He sends you his greetings. We were talking bit together about you and he came out, without my having raised this at all, with a remark about your immense facility as an algebraist. This has been somewhat my own feeling but runs dead counter to this remark of yours. I wonder if I could vex you by asking you to say a little bit more about what you'd had in mind.

Dirac:

Well, I'm not altogether sure what is meant by 'algebraist'. If it means some- one who simply carries through masses of algebraic calculation without picturing what the equations mean, then I'm just no good at it. All this modern work about dispersion relations and reggi-poles and things like that I find very difficult to follow. It doesn't impress me strongly at all because I don't see the geometrical connections.

Kuhn:

I would think of your peculiar q-number manipulations, for example, as being algebraic rather than geometric.

Dirac:

Yes, but I only used them in an elementary way. Perhaps I didn't tell you that I kept up my connection with geometry some time after I came to Cambridge. There was a Professor Baker, a professor of geometry, who used to give tea parties on Saturday afternoons to people who were keen on geometry; after the tea someone would give a talk on some recent research work on none geometrical subject. I went to those tea-parties and absorbed quite a lot of geometry then. I talked once or twice myself. I remember I worked out a new method in projective geometry and gave a talk about that at one of these meetings. I never published this method. Well, that's a good deal about working with the geometry of four or more dimensions. Four dimensions were very popular then for the geometrists to work with. It was all done with the notions of projective geometry rather than metrical geometry. So I became very familiar with that

kind of mathematics in that way. I've found it useful since then in understanding the relations which you have in Minkowski space. You can picture all the directions in Minkowski space as the points in a three-dimensional projective space. The relationships between vectors, null-vectors and so on — and you get at once just the relationships between points in a three-dimensional vector space. I always used these geometrical ideas for getting clear notions about relationships in relativity although I didn't refer to them in my published works.

Kuhn:

Did this also give you techniques that were relevant to your approach to quantum theory whether relativistic or not?

Dirac:

No. It doesn't connect at all with non-commutative algebra.

Kuhn:

It is all right to think of that as being algebraic.

Dirac:

Yes, yes. But I don't think you'll find any heavy algebra in any of my work.

Kuhn:

No. There is an awful lot of algebraic sense, though it's hard to know how to put that more precisely; but I think with many people who find the approach to a multiplication relation more general than XY , where these cannot be thought of as numbers, it suddenly gets away from them as a subject matter. You've clearly made this exception with ease and facility, at a time, I take it, when it wasn't the standard thing to do.

Dirac:

Yes, I suppose. Non-commutative algebra was a rather strange idea in those days, although it shouldn't have been because of quaternions. Hello, come on in. (Someone enters room)

Kuhn:

I think we were talking about the transition from engineering to math and I think it might be appropriate to go on now to the transition from math to physics.

Dirac:

When 'I came to Cambridge, yes.

Kuhn:

I take it that although you had had a year in which you turned out to be in applied math, this wasn't necessarily to be a career with the subject matter that you ultimately went on with.

Dirac:

Not necessarily, no.

Kuhn:

Did you think of yourself as going to Cambridge to do applied, math?

Dirac:

It was already when I was finishing my engineering course, my father sent me to Cambridge to try for a scholarship. Actually, it was too late when he thought of it to try for the honorary scholarships because the examinations for the honorary scholarships are in December but St. John's College offered some exhibitions for which the examinations were held in June. I went and took that examination and they offered me an exhibition of 70 pounds a year. I wasn't able to get any other money to supplement it and it wasn't enough to come to Cambridge with. That's why I stayed on in Bristol. But after the two years in Bristol I was awarded a government grant from the Department of Scientific and Industrial Research and with the help of that I was able to come to Cambridge.

Kuhn:

Could you also take up the previous scholarship?

Dirac:

Yes. The previous exhibition. They allowed me to take it up two years later. It was still a very small amount of money, but just enough to manage with.

Kuhn:

Was your father able to help you at all?

Dirac:

He didn't seem to want to anyway and I did get by entirely, in this way, independently of my father. He left quite a bit when he died, but at that time he didn't think he could afford it.

Kuhn:

If you had gone to Cambridge immediately after finishing your engineering course, would you have gone into applied mathematics or mathematics?

Dirac:

To begin with, one doesn't specialize. I would have had to specialize later. don't know how it would have turned out.

Kuhn:

Well, since you had already finished your engineering work, you'd have had some idea of going to study something.

Dirac:

It would have been mathematics, yes.

Kuhn:

Well, in that sense there really wasn't as much accident involved as I had thought before in the transition from engineering to mathematics in the graduate school at Bristol.

Dirac:

That was made quite independently of my having won this exhibition in Cambridge.

Kuhn:

But at least you would have gone on to do mathematics at Cambridge. Well, coming back now to when you actually do go there in '23, you'd had a good deal of mathematics in the meantime. Was it the sort of mathematics by that time that meant that you were thinking of a career in teaching?

Dirac:

I didn't like the idea of being a school-teacher anyway.

Kuhn:

Did that seem the likely outcome?

Dirac:

Well, if I wasn't very good, it might have been the only possibility. I didn't like the idea of becoming a schoolteacher.

Kuhn:

The other alternative was university teaching?

Dirac:

Yes. I didn't know whether I'd be good enough for that.

Kuhn:

Clearly you had impressed an awful lot of people quite a lot; was there real doubt in your own mind about the likely outcome of this?

Dirac:

I was really very ignorant of the world. I had done well on examination but that was all. I had no idea what the standard was in Cambridge.

Kuhn:

You went there more or less with an eye to continuing in more advanced subjects in mathematics?

Dirac:

To do research. The government grant that I got was specifically for doing research

Kuhn:

You didn't have to have a particular research project.

Dirac:

No, no.

Kuhn:

It was research in mathematics, but it might have been anywhere in mathematics.

Dirac:

Yes, yes. So I came to Cambridge as a research student and not as an undergraduate. That was why I was able to come without knowing Latin. If I had gone to Cambridge earlier as an undergraduate I would have had to suddenly learn Latin.

Kuhn:

Did you attempt to pick up research problems virtually on arrival, or did you go to lectures and read at first?

Dirac:

Well, Fowler was appointed as my supervisor.

Kuhn:

This was immediately on arrival.

Dirac:

Yes. I think perhaps I first went to see Cunningham. I had seen Cunningham previously when I came for the examination for the exhibition. I was rather hoping that Cunningham would be my supervisor. He worked in electromagnetic theory and I thought that was an interesting subject. Cunningham couldn't take me. I don't quite know why. I was assigned to Fowler and his interest was statistical mechanics and I thought that to be a relatively ugly subject.

Kuhn:

Had you looked at it at all before?

Dirac:

Not really, no. Cunningham is still alive, by the way; if you'd like to ask him any questions you probably could. He lives here in Cambridge. Well, Fowler set me to think

about this question of dissociation under a temperature gradient.

Kuhn:

At the very start?

Dirac:

Pretty near the beginning, yes. He also gave me things to read and lectures to go to. Fowler himself was giving lectures on quantum theory, I believe. It was then that I first learned about the Bohr atom. I was quite surprised to find that atomic theory had developed to such a large extent.

Kuhn:

Do you still have notes on those lectures?

Dirac:

It's possible that I have them. I don't remember where they are; I could look for them.

Kuhn:

It would be terribly helpful to have a precise idea. at what level, over what illustrations and subject matter Fowler was actually lecturing at that quite key point. So if you do have those I think we would like very much to microfilm them.

Dirac:

I also went to Cunningham's lectures. He was lecturing on classical electrodynamics. I went to lectures on thermodynamics. I believe that was by Newman. Newman gave some lectures on relativity and so on. He is now in Manchester. I went to the Colloquia at the Cavendish. We sometimes had visitors from foreign places. Bohr came and lectured sometimes. Of course, I was very happy to meet Bohr. I was also very happy to meet Eddington, because I had heard about him. so much in my engineering student days. Eddington was the great man then. He brought relativity to England, and he led that eclipse expedition to check the Einstein effect in 1919. That aroused tremendous public interest.

Kuhn:

Did you go to lectures of Eddington at that time, or courses of lectures?

Dirac:

I can't remember whether I did or not. I don't remember now.

Kuhn:

Well, there are all sorts of subjects coming in quickly now that are really new to you. I mean the electromagnetic theory ties somewhat to your previous engineering training, but must go a great deal beyond it. Statistical mechanics is a subject which was perhaps quite new.

Dirac:

Yes, yes. Lennard-Jones was in Cambridge at the time. I learned quite a bit from him. Well, I got introduced to the Boltzmann equation for the first time. This was a starting point for statistical mechanics. I also learned some astrophysics from Milne. During one term when Fowler went to Copenhagen I was appointed my supervisor and he set me on that astrophysical problem on which I wrote a little paper.

Kuhn:

Unless my impression is wrong, when you came to Cambridge you start on more advanced work, but really the whole subject matter broadens out.

Dirac:

Yes.

Kuhn:

Now, at least from the point of view of the American vocabulary, you are doing physics as you had not previously been doing physics. Many of these subjects are applied mathematics or astronomy at Cambridge — this is not the natural way to describe it.

Dirac:

Well, the electromagnetic theory was really physics.

Kuhn:

Could you have had many of these same subjects at Bristol or did they simply not exist there?

Dirac:

Statistical mechanics and thermodynamics I don't think existed at Bristol.

Kuhn:

Electromagnetic theory probably did?

Dirac:

Yes, but not the waves. Not electromagnetic waves. I learned a little about the lagrangian and a little about the Hamiltonian.

Kuhn:

But that existed in the sense of potential theory problems.

Dirac:

Yes, quite a lot of potential theory problems.

Kuhn:

Well, I would be grateful if you could look at some point and see what lectures you may have notes on. It would help to pin down content of lectures more specifically and there may very well be a few of those that we'd like to microfilm.

Dirac:

I think my notes are pretty rough. In any case, I didn't absorb very much from lectures. I got a more thorough understanding from private reading than from lectures.

Kuhn:

Do you remember again books you read in this period?

Dirac:

Sommerfeld was the main book. Sommerfeld's Atombau and Spectralinien.

Kuhn:

You read that in the German. There was an English translation-

Dirac:

It was an English translation, but it was rather a bad one. I think I read it in the German. I had read some German in school and it turned out to be rather inadequate. Still, with the help of a dictionary, I was able to plod along and gradually got more fluent with it.

Kuhn:

But Fowler's statistical mechanics was not out yet.

Dirac:

Oh, no. Not till much later. But I read a good many papers and periodicals.

Kuhn:

What periodicals would you have been likely to follow at that point?

Dirac:

The Proceedings of the Royal Society, and the Zeitschrift für Physik, that was the main physics journal in those days. Annalen der Physik I recall. I don't know whether the Physical Review existed then. It wasn't very important.

Kuhn:

Yes, it did exist, but I find almost nobody who would have been likely to be reading it in Europe. It had existed for a while, but was not an international journal.

Dirac:

Well, the Zeitschrift für Physik was the main physics journal in those days. And it was from reading these papers that I got ideas for problems in research work.

Kuhn:

Some of those also must have come from teachers. That is, Fowler you say put you on right at the start.

Dirac:

He definitely put me on this first problem as soon as I arrived.

Kuhn:

Is it that second one that is the relativity dynamics of a particle?

Dirac:

That's just a very short note. That was suggested by something that Eddington had written and Fowler had talked about. Seems that I could put things a little more clearly than he had them. Yes, it was in Eddington's book.

Kuhn:

That came out of the book you think rather than from taking a course with him?

Dirac:

I think it came out of a book, yes.

Kuhn:

The paper you do on the Doppler principle and the Bohr condition that may well come out of reading.

Dirac:

Yes, what was the title of that paper?

Kuhn:

"Note on the Doppler Principle and the Bohr Frequency Condition."

Dirac:

I expect it did. I've forgotten that paper. Was that in the Cambridge

Kuhn:

Yes. That's the

Dirac:

I think that did come from private reading.

Kuhn:

Well certainly there is a relevant paper of Bohr's that you point to that's In the supplement to the Cambridge Proceedings.

Dirac:

Yes, I think this sort of work came just from reading things and noting that it's possible to improve what someone else has written

Kuhn:

In connection with your next paper on the detailed balancing in the Proceedings of the Philosophical Society [Royal Society A 106], I'd like to ask whether by any chance you knew of de Broglie's work in this period.

Dirac:

1924, was it?

Kuhn:

Yes.

Dirac:

I did know about it and I wasn't strongly impressed by it because it seemed to me to be a mathematical curiosity. I didn't take these waves as something physically real in the way that Schrodinger did. It never occurred to me to try to generalize it for particles which are not moving uniformly. I looked upon it just as a mathematical curiosity.

Kuhn:

Do you remember how you knew of it, because there are lots of people who were totally unaware that it existed in this period.

Dirac:

I think de Broglie had published something about it. I had read about it.

Kuhn:

Do you remember where?

Dirac:

I don't remember where, no.

Kuhn:

The thesis itself, although it was published in full in French, was not often read; there was however an article in the Phil. Mag., that summarized it.

Dirac:

I probably got it from that.

Kuhn:

Fowler was the one who transmitted the article to the Phil. Mag., I think it was Ellis that was responsible for its being there in the first place because he had been at Maurice de Broglie's lab. But I wonder whether you had ever heard Fowler speak of it or remembered other conversations about it.

Dirac:

No, I don't remember Fowler speaking about it. I don't think it's the sort of thing that would impress Fowler. Fowler's interests were in statistical mechanics and the Bohr orbits also.

Kuhn:

Well, of course this tied to both of those, you know. That is, de Broglie had a derivation for the quantum condition or for the Bohr orbit in terms of the interference and he also got some quite interesting statistical results. So I'm not sure whether those were in the article or not.

Dirac:

I didn't know he had any statistical results. I think it was just speculative what he said about the possibility of connecting these waves with quantum condition. It was just a speculation. I don't think people took it seriously. Nobody except Schrodinger took it seriously.

Kuhn:

Einstein took it fairly seriously, I think. There are an awful lot of people who simply

didn't know it existed. There was nothing equivalent to the Phil article printed in any German journal that I know of. So by and large it was unknown to people who didn't follow the French literature. I was really led to ask you this question in the first place because of the one passage in the paper on detailed balancing in which I supposed probably you were speaking directly to the de Broglie paper.

Dirac:

I don't think I referred to it in any [part of the paper].

Kuhn:

You certainly didn't cite it. It was this on page It's that paragraph. [End of § 5. "For the discussion of equilibrium problems, quanta of radiation cannot be regarded as very small particles of matter moving with very nearly the speed of light."]

Dirac:

Well, so far as concerns the light quanta and the electromagnetic waves, de Broglie's ideas were common to everybody but the idea of extending them to other particles with non-zero rest mass was a new thing, which people, did not take seriously.

Kuhn:

But de Broglie, so far as I know, is the only one who in talking about light quanta talks of them as having a very small but finite rest mass and a velocity correspondingly just a little bit less than the velocity of light. That is, most people simply give them the velocity of light and zero mass. De Broglie from the start talks of having a very finite mass and a velocity slightly less than the velocity of light and you phrase this point on what I agree is in general a well-known idea of light quanta. But this phraseology strongly suggests to me that maybe you're thinking of de Broglie's version of it.

Dirac:

I'd better think this over and see if I can remember any more about it.

Kuhn:

O.K.

Dirac:

We ought to go and have our tea now.